Andrea Brilli

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Sapienza, University of Rome, Department of Computer, Control and Management Engineering "Antonio Ruberti"

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Education

Sapienza, University of Rome, Operations Research — Rome, Italy, 2021 – 2025
- Curriculum: Operations Research
- Thesis: Derivative-Free Optimization: worst-case complexity for Line-Search methods and a mixed Penalty-Barrier approach
- ABRO (Automatic control, Bioengineering and Operations Research)

Sapienza, University of Rome, Management Engineering — Rome, Italy, 2019 – 2021
- Curriculum: Decision Models
- Thesis: An interior penalty method for black-box constrained optimization

Sapienza, University of Rome, Management Engineering — Rome, Italy, 2015 – 2019

Experience

Sapienza University of Rome, PostDoc — Rome, Italy, 2024 – present, 2 years

Department of Computer, Control and Management Engineering. Funded by PRIN (NextGenerationEU funds) for hydrodynamic optimization study on biomimetic and innovative hydrodynamic configurations for autonomous underwater gliders (AUG), specifically designed for ocean surveying.
- Support to PhD students on advanced techniques for training neural networks and derivative-free techniques
- Collaboration with INESC MN (Lisbon, Portugal) for optimization of structural design parameters of metamaterials
- Collaboration with NOVA University of Lisbon on techniques to approximate the Pareto Front of constrained multiobjective optimization problems
- Collaboration with Polytechnique Montréal to study nonsmooth constrained optimization problems
- Collaboration with University of Florence to enhance performance of Augmented Lagrangian techniques

European Space Agency - ESRIN - ϕ-lab, Research Visiting — ESRIN, Italy, 2026 – present, 1 year

Collaboration to develop denoising techniques for satellite data striping noise.

Polytechnique Montréal, Ph.D. Visiting Researcher — Montréal, Canada, 2024 – 2024, 1 year

Six months of collaboration with Professor Sébastien Le Digabel and Professor Youssef Diouane to enhance the performance of NOMAD solver.

Universidade NOVA de Lisboa, Ph.D. Visiting Researcher — Lisbon, Portugal, 2022 – 2023, 1 year

One year of collaboration with Professor Ana Luísa Custódio and Ph.D. Everton Jose da Silva to enhance the performance of the SID-PSM algorithm.

Teaching

Sapienza University of Rome, Mathematical Programming — Rome, Italy, Sept 2025 – Jan 2026
- Introduction to multivariate calculus and mathematical optimization
- Bachelor degree in Computer Science Engineering

Awards

AIROYoung Dissertation Award Sept 2025
Yearly award for the best doctoral thesis in Operations Research defended in Italy.
AIROYoung (Italian Operations Research Society)

Publications

Complexity results and active-set identification of a derivative-free method for bound-constrained problems
Accepted for publication in JOTA. Analysis of complexity and active-set identification for derivative-free optimization with bound constraints.
A. Brilli, A. Cristofari, G. Liuzzi, S. Lucidi
arxiv.org/abs/2402.10801

A penalty-interior point method combined with mads for equality and inequality constrained optimization
Novel constraint-handling method combining penalty and interior point approaches with MADS algorithm.
C. Audet, A. Brilli, Y. Diouane, S. Le Digabel, E. Silva, C. Tribes
arxiv.org/abs/2601.20811

Tunable ito–metal plasmonic metamaterial channel for tailored sensing: A simulation-driven approach
Optimization-driven design of plasmonic metamaterial sensors for tailored sensing applications.
C. Alfisi, A. Brilli, H. Terças, S. Cardoso
doi.org/10.1063/5.0297834

An interior point method for nonlinear constrained derivative-free optimization
Novel interior point approach for solving general nonlinear constrained problems without derivatives.
A. Brilli, G. Liuzzi, S. Lucidi
doi.org/10.1080/10556788.2025.2453110

Nonlinear derivative-free constrained optimization with a mixed penalty-logarithmic barrier approach and direct search
Mixed penalty-barrier method combined with direct search for general constrained derivative-free optimization.
A. Brilli, A. L. Custódio, G. Liuzzi, E. J. Silva
arxiv.org/abs/2407.21634

Worst case complexity bounds for linesearch-type derivative-free algorithms
Theoretical complexity analysis for linesearch-based derivative-free optimization methods.
A. Brilli, M. Kimiaiei, G. Liuzzi, S. Lucidi
doi.org/10.1007/s10957-024-02519-x

Skills

Optimization

Programming

Software Development

Languages

Italian
Native speaker

English
Fluent

Portuguese
Basic

Spanish
Basic

Certificates

Deep Learning Specialization 2020

Data-driven Astronomy 2020

CyberChallenge 2020 2020
Training programme organized by CINI in collaboration with the Ministry of Defence and SISR.

Projects

DFL - Derivative-Free Library 2021 – present
Collaborator for the implementation of algorithms in Python within the DFL library. Author of LOG-DFL.

LOG-DS 2023 – present
Co-author of the LOG-DS algorithm, a direct search method for general nonlinear constrained problems implemented in Matlab.

NOMAD 2024 – present
A Blackbox Optimization Software. Supported the implementation of new constraint-handling strategies within the package.

References

Professor Giampaolo Liuzzi

Professor Stefano Lucidi

Professor Ana Luísa Custódio

Professor Sébastien Le Digabel

Professor Youssef Diouane